

MDA: Multiple Decentralized Anchors for Hiding Communication Information

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Abstract—This paper presents a novel network architecture based on The Onion Routing (TOR) network and BitTorrent concepts to advance privacy and anonymity in communication networks. In the traditional TOR network, when the transmitter sends a message, any external intruder or the first proxy knows that a message has been sent from a specific location or IP address. The same issue is on the receiver side. This can reveal some privacy about who using the network, the time of activity, and the density of communications. Another issue with the services that provide end-to-end encryption is that the powerful entities can force the service's owner to deliver their encryption keys and other related data that can reveal information about the communications within the network. Therefore, the new architecture decentralizes the network entirely in which nodes are the only fundamental component of the network. The nodes themselves act as hops and control the rules of the network by using a distributed hash table (DHT). There are no proxy servers working as hops in TOR networks or servers to hold DHTs as trackers in the BitTorrent network. The nodes act as an end client and a hop server for other nodes that want to join, leave, and communicate. The actual implementation of the MDA network is under development for peer-to-peer (P2P) messaging. Further development can be made for web browsing, file sharing and video streaming.

Index Terms—Decentralized Networks, Anonymity, TOR network, Onion routing, P2P networks, BitTorrent network, Internet network privacy, serverless networks

I. INTRODUCTION

In today communication networks e.g. Internet, cellular and wireless sensor networks, the privacy and security of these networks attract more concerns, especially, with the introduction of new technologies e.g. Internet of Things (IoT), blockchain and cloud servers [1], [2]. In 5G cellular network and beyond the number of devices will be 1000x the present number of devices where the high throughput and low latency communication will be deployed widely [3]. Recent research introduces many terminologies, infrastructures and network protocols to enhance the security and keep the end user privacy [4]. However, many services e.g. social media, mobile and Internet applications reveal and analysis the end user information to improve their performance or provide this data as a product for other entities. This can lead to data and traffic analysis by intruder or observer which downgrade the privacy dramatically. Although the traffic can not be revealed itself due to the usage of encryption algorithms e.g. Public and Private

keys, the activity of end users at the edge of the network can link them together to predict who communicates with whom, who are using a private communication service, who searches for what, their locations, time of activities, their connected devices, and other activities outside the private communication service [5]. Despite of the contents of the messages between the transmitter and receiver, their identities and activities still an important part of the communication information and its privacy. Therefore, the proposition of new anonymity concepts that hide the communicating parties and related information still in need [6].

The most well-known anonymous network is TOR networks which depends on a series of relay nodes between the transmitter and receiver where the message delivered through a secure tunnel between each two successive relay nodes. Then, the next node delivers the message to its successive relay node without knowing about other nodes involved. This series called *circuit* where the transmitter ask an entry node to builds it from the hidden routers within the TOR network to complete layers of encryption to cover the original message like onion layers. The entry node knows the entire circuit [7]. The real message is in the core and covered by layers of encryption. Each layer is related only to one relay node in the closed circuit that already have been built at the initiation of the communication session. When the message passes one relay node within the circuit, one layer recovered until the message reach the receiver [8]. TOR networks have been designated to provide a low latency privacy service that can be used in a streaming application e.g. web browsing. However, this vulnerable for partially observation and traffic analysis by a Global Passive Adversary. Various studies review cases of de-anonymisation attacks and privacy revealing. As a result, more research still needed to enhance the anonymity and privacy of TOR networks [9]–[11].

The most well-known peer-to-peer (P2P) decentralized network is the BitTorrent network which is basically a file transfer protocol (FTP) used to spread copies of files widely without using any servers to store the file itself. The nodes that store the file or parts of the file called seeds. When one node demands that file, it needs a list of the nodes that have portions of the file and download it by parts. This list called Hash table which

contains the hash ID and IP address of each node. Finally, the node gather all parts in one complete file ready for use [12]–[14].

Crowd theory (Crowd psychology) or crowds is simply the study of the actions, flow and characteristics of many people happened to be in the same place, time or any other criteria e.g. share same network, under same circumstances [15], [16]. Any entity can hide himself either by building many strong defense layers or walls equipped with defense tools to protect herself against various attacks or by anonymizing himself as possible within the crowds using their characteristics and behaviors to be an identical copy of any other entity within the crowd. This is called going with the crowd (GWC) which means that one discards his decisions to be one of the crowd and follow their overall behavior [17]. There is a lot of research on the behavior of the people within crowds and the dynamics of crowds [18].

The theoretical concept of the proposed anonymity network, in general, is a mixture of TOR network (Onion concept), BitTorrent network (Hash table concept) where both of them are in IP-based networks, and the crowd theory. The aims to be achieved:

- 1) Achieve the complete theoretical anonymity where the transmitter and receiver can not be linked together and normalizing their activity to be identical as all other nodes within the network. This can not be revealed even by a Global adversary where all the links and activities are monitored.
- 2) Anonymity the single end node within the network against intruders with less complex protocols to preserve the privacy while not degrading the communication performance. This requirement implies that the new network should run at low consumption of both user and network resources as many devices have highly restricted resources.
- 3) Provide a simple and practical design that can be deployed widely over any kind of communication network. As the anonymity requires the hide within the crowds. A practical anonymity network should be used from any device without any complication in installation and configuration.

The contributions of this paper are lists below:

- 1) A practical implementation of a decentralized anonymity network based on the basic principle of the TOR and BitTorrent protocols. The only component in the MDA network is the node which acts as an end node and as a hop server for other nodes in its networks. The MDA network is a complete developed architecture that can be implemented as a protocol under any type of communication service.
- 2) A seed file called mdar that is basically a list of nodes within the same small MDA network and its rules. Privately or publicly mdar file can be shared and updated between nodes in the same network to use it as a list of proxies for communications between members of the network.

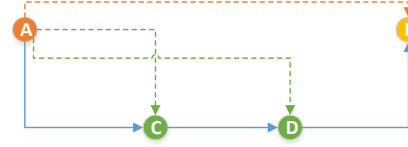


Fig. 1. An illustration of TOR network with one transmitter, one receiver, two hops and layers of encryption between the transmitter (node A) and the receiver (node B).

- 3) Every node sends random number of fake messages frequently to hide its real activity. The fake message is identical to the real one in both the size of the message and the routing process. The only different between two fake and real message is that the real message is readable and meaningful at the receiver side.

The rest of the paper includes threat models discussed in Section II. Section III explains MDA's design model for joining, communicating and leaving the network. In Section IV, an analysis of the threads is presented and how MDA network deals with them. Finally, Section V concludes the work and presents further improvement and open problems.

II. THREAT MODELS

As in TOR network, when transmitter wants to contact the receiver, he chooses multiple nodes as hops and make a connection between each three consecutive nodes. Each node knows only the previous and next nodes in the chain. Then, a message is encapsulated with layers of encryption corresponding to the hops in the onion routing. At each hop, a layer is revealed until it reaches the receiver which decrypts the message to the plaintext. Fig. 1 shows a simple overview of the TOR network where the transmitter (node A) firstly initiates the circuit and connect each node with its predecessor and successor nodes. Then, the transmitter starts the communication. The main threat to the TOR network is the Global Passive Adversary who can monitor most of the network links, who is using it and when. This kind of surveillance exposes the privacy and can with some probability knows the transmitter and receiver identities geographically by revealing of the hops of the route that links the transmitter with the receiver. Also, by monitoring the activity of each user on the network, this can give information about who get online on the same time or the increase of activity at a specific website when a target uses the network. Although the message contents can not be revealed due to encryption, previously collected information about the communicated users can reveal with some probability the purpose of communication.

The job to be done by the ideal anonymity network is to hide the communication components that include transmitter, receiver, message, environment, and related user activity by normalizing them at all network nodes to be identical. Thus, any intruder can not know who communicate with whom or are they communicating at all. The communication environment

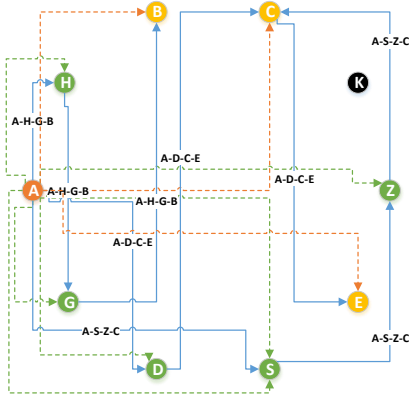


Fig. 2. An illustration of MDA network with one transmitter (node A), three receivers (node B, C, and E) and two hops on each connection. The three messages to deliver can be real or fake messages. The control messages are shown in dash lines which have the same characteristics of real and fake messages.

means the devices, protocols, type of message, and dense of communication. User activity is related to the usage of the network in both time and geographical location e.g. when, from where, and how many real connections the user makes. MDA design tries to cover the nodes within the crowds and make the real connections undetectable by the behavior of the users themselves. It is not a protection against the attacks on the end users' devices.

III. THE MDA NETWORK DESIGN

The design takes the advantages of TOR network, BitTorrent network and Crowds theory as follows: 1.

- 1) From TOR network, the idea of multiple relays between the transmitter and receivers are chosen by the transmitter. Then, the transmitter makes the route and initiates the connection. These relays are chosen from the hidden servers that belongs to entities with different interests. In MDA network, the relays are chosen from the other nodes within the network.
- 2) From Torrent network, the idea of Hash tables that include the most basic information about the nodes related to the subject of interest e.g. movie file. These nodes are called the seeds. Any node can make be a seed and include itself in the torrent file. In MDA network, a group of nodes lists themselves in mdar file to be seeds for the network. Therefore, any node initiates the connection by choosing randomly from the nodes included in the node list as seeds.
- 3) From Crowds theory, the idea of anonymous node which has exactly same characteristics similar to other nodes. For instance, in train station, for an observer, all persons have basic information that he can not differentiate between them. Assume that all persons in that station wear same clothes, shoes and use same kind of movements to be completely identical. From the observer point of

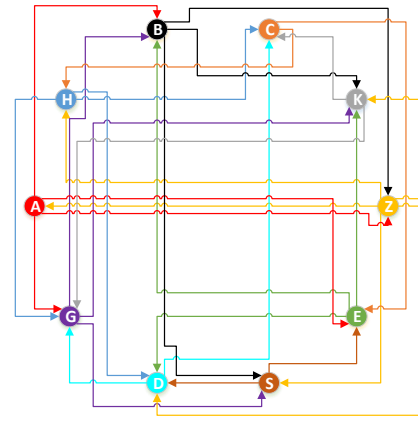


Fig. 3. Overview of the MDA network (10 nodes) at any given time where each transmitter chooses the number of receivers randomly between $\{1,5\}$ from the node list of this network. These links include initiating control, fake, or real messages. Besides, links for re-transmitting any relayed messages.

view, it will be much harder to distinguish an individual from others. Another example which is more obvious, the swarm colonies e.g. ant, bee. In MDA network, every node is an identical seed of rest nodes. Also, the messages between nodes are identical. Further, the activity is normalized to show same activities at all nodes.

In this section, the node internal architecture, the MDA network architecture are presented. Also, the different types of links within the network are described.

A. Node Implementation

The normal client to server connection is used to simplify the design and implementation of the MDA network. As the node is the only foundation, it has both client and server functions. Therefore, every node has two main parts: Node client and Node server. The Node client acts as an end user protocol for higher level applications. Its main jobs are sending real requests made by the user and receiving replies. Node server acts as a hop. Its functions include initiating fake messages, receiving control messages, completing connections with other nodes that want to initiate a connection with this node as a receiver either for a real or fake messages. Finally, re-transmitting relayed messages.

In MDA network, there are three different connections including internal, control and messaging. Internal connection is the link between the Node server and Node client. Control connection is for networking like updating hash tables periodically so all nodes know any updates e.g. new and deleted nodes, non-active and active nodes. The messaging connection is for exchanging all kind of messages e.g. real and fake.

1) *Node Client*: The Node client has a simple application layer which has a user interface for messaging and other related functions e.g. save messages, friend lists, open and end a messaging windows, ..., etc. Also, it is responsible for

initiating, starting and ending the real messages with the Node server at the receiver side. In the case of being a real receiver, the Node server sends internal message with the initiating information to start the connection. The Node client completes the task of initiating the connection as a receiver and start exchanging messages with the transmitter.

2) *Node Server*: Node server is the automated part which control the network with other nodes and stands for any incoming messages. If the received messages belongs to a real connection or tries to initiate a new connection, Node server sends to the Node client. Otherwise, the Node server handles both control messages related to the network, initiating fake messages, and relaying messages from other nodes.

TABLE I
DESCRIPTION OF DIFFERENT TYPES OF MESSAGES EXCHANGED WITHIN
MDA NETWORK.

Message type	Description	Routing	Behavior
Real	A real message send after the connection has been initiated by intended user with known receiver within the MDA network.	Onion Routing.	Active when the user sends or replies within a pre-established secure MDA connection.
Fake	A randomly generated message send after a random connection has been initiated automatically by any node with a random receiver.	Onion Routing.	Active as the user online and frequently initiates random number of connections.
Control	A control message e.g. initiate connection, which has the same characteristics like other real and fake messages.	Direct between two nodes.	Frequently updates the node list and other network services.

B. Network Implementation

Any node that needs to contact another node will select a random set of relays from the node list of the network that both transmitter and receiver are in. Any node can join multiple networks and start connections with nodes listed on them. However, both the transmitter and receiver should be on the same network and have the same node list to communicate. Then the transmitter constructs an encrypted connection between each successive nodes on the route. Thus, each node knows the previous and next node but not the entire route. When a node receives a message, the node decides, if it is a control message, then the node deals with it accordingly. If the message is from previously initiated route, then the node forwards the message to the next relay node. If the node is

the final receiver of the message, then the node decides if the message is fake or not. If the message is fake, the message will be discarded. Otherwise, it is real and the node decrypts and shows it accordingly.

This network is built on IP-based network. Thus, the transmitter knows the receiver by his IP address and his network ID. For other kind of networks, it will use their underlying physical and network layer. Also, for security and privacy, all messages even the fake will be encrypted using traditional two-keys encryption algorithm e.g. RSA. This will give each connection between any two nodes in the network the same characteristic for any other connection.

1) *Creating The Network*: The network depends on the seeds to build it up. Seed nodes can be any end users with similar interests or voluntary individuals or entities. Any number of seeds can build up an MDA network by making a mdar file containing them in a list of nodes as seeds for the new network. Then adding the file to their client to get updated frequently as any changes happened to the list. For instance, a node changing his IP address, leaving the network or not active for long time. These seeds have a responsibility to put some rules for the network. These rules are agreed and shared on the first place. For instance, no further nodes can join, new nodes should get acceptance from one seed, or the network is public and any node can join directly. These rules can be extended to ensure that any new node is not intruder and will act normally.

2) *Joining The Network*: For the node to choose the relay nodes, it has to join and know a MDA network's node list. Here we can use of Torrent Hash list where each node has an ID and lists its IP and PORT in the node list. To join the network, the new node should ask any seed in the network with a joining request or the seeds whose responsibility is to monitor the network rules. The seed client can manage the joining request manually by having a human approval or automatically by replying directly to the new node. The reply message will have a confirm and the updated node list along with a broadcast to other nodes with the new updated node list. In every reception of the node list, a filter is happened to update the list correctly with all new added and old removed members. All approved by multiple seed nodes. Further, seed nodes can be extended to be any nodes in the network. However, If a node G tries to establish a communication with node B while node G is not in any network with node B, the connection will be refused.

3) *MDA Routing*: The MDA network works as an anonymity layer for its members. At different time frames, the node sends random number of messages including control, real, and fake messages. All types of messages are identical to cover on the both the control and real ones. Even if there are no real messages, the node generates and initiates fake messages. The routing for both real and fake messages follows the same as in TOR network. The transmitter establishes a key exchanges with all the node all along the route. Then, it sends the real message.

IV. DISCUSSION

In this section, possible threats to the MDA network, and how the MDA network preserve anonymity of the users against them are discussed. The most of threats to the MDA network comes from the fact that the IPs of the nodes are available in the node list. As users of the MDA network is known for any intruder with the node list, there are some actions that can be made with the nodes IDs and IPs. Firstly, he can introduce some nodes in the network, that can normally interact and participate in the MDA network which can reveal some connections within the network. These connections either a node want to use the intruder node as a hop server or its a final destination for a fake message. In both cases, the intruder only knowledge is that the node want to use the network. However, he can not reveal the identity of the real receiver in case that a real message is being transferred through his hop server because of the following reasons. The message is encrypted with layers, it is not known if the message is real or fake, and it is not known if the successive node is the final receiver or there are more nodes in the series. In case that the intruder node is a final destination of a fake message, the full discover of the series is usefulness.

Another thread is that the intruder can analyze the network connections of the target node for all possible connections through the MDA network. The possible connections that be revealed are control messages that related to the MDA network itself which is available for all nodes using the same MDA network. Also, the control messages related to initiating a new connection. The intruder can know that a communication with another node is going. However, he can not know which one of the nodes is the final destination or is the message to be transferred is a real or fake message. For some time, the intruder can know in a certain time, a link is fixed while other links destroyed after some time varies according to the number of hops in each link and network connectivity. This can be resolved by two ways, either by fixing a certain time for all connections. However, this limit the time for the real connection but give the best anonymity. The other option is to re-construct the whole links with some nodes every while to anonymize the real link within these links. The potential risk is that all the final nodes in the fake links are belongs to the Intruder which is difficult with highly restricted rules of the groups that include the nodes. This open the research for the responsibility of the crowds to act as one and be aware of possible vulnerabilities instead of keeping the full responsibility on the end user.

Nodes list is designed based on the degree of anonymity which can be restricted to highly trusted users within a private group. This can not lead to any leakage of the list of nodes. However, the intruder can analyze the traffic of a target node which can lead to a number of nodes without knowing the type of messages or revealing the route for a target real link. This can be more complicated by using multiple groups and hence, the intruder can not know which node belong to which network or how many networks the target node is a member

of.

Another novel idea is to not keeping the group members for a while. This means that not only the messages has a limited life time but also the groups. This is intended against a powerful adversary who can analyze the whole network and target all the nodes that uses the MDA network and its groups. Analyzing the whole network for an enough time can lead to a building a knowledge about which node belong to which group and thus narrowing the scope of analyze. However, keeping changes increase the complexity and add more layers to the anonymity of the MDA network. Another layer of anonymity is by using a hop node as a proxy or entry point to the MDA network and changing it every certain time.

The MDA is not invulnerable against all possible attacks and this discussion does not cover all these attacks. However, it is listing the possible attacks from the architecture point of view. Any highly organized entities that can include other services like Virtual private network (VPN) and proxies which can increase the complexity for a powerful global adversary. However, in this paper we just introduce the fundamental architecture of the MDA network.

V. CONCLUSIONS

In this paper, the MDA anonymity network has been introduced for real time full duplex communications. The MDA network is based on the TOR network, BitTorrent network, and Crowds theory. MDA network does not rely on any middle servers or any hidden services. It is only relying on the end user as a node to participate in the network as an end client and as a hop server for other nodes. The conceptual design is implemented around creating crowds of identical individuals and hiding the the information of users within the crowds and their communication. In TOR network the entry node knows the route including the destination while in MDA network only the transmitter knows the all the nodes on the circuit and the final destination. Although, the implementation has been done for IP-based networks. The concepts of MDA network can be extended to other types of communication networks e.g. wireless sensor networks, and ad-hoc networks. The anonymity of entities within the MDA network would be effective within participation of huge number of individuals or end users represented by nodes in the MDA network architecture. The MDA network is intended to provide a anonymity while not using any pre-allocated resources from any organization or entities and reserve the sole responsibility for privacy and anonymity on the collaborative work of the crowds. Although this may lead to failure if the crowds decide not to follow the MDA rules predetermined by groups created within the MDA network. Further research and development will be done to extend the use of MDA protocols into other types of communications. Also, the integration of MDA networks into various types of connected devices that want to send and receive anonymously. Finally, developing the idea to cover all the aspects of MDA network and future unforeseeable vulnerabilities.

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